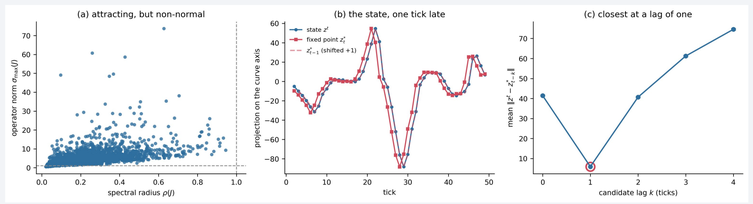
Riding the Fixed-Point Curve

Image classification in the Continuous Thought Machine

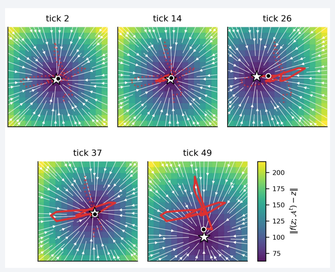
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Attracting, but non-normal



- **Attracting**: ρ(J) < 1 at every fixed point found (3055 of 3072; median 0.15); 3456 restarts from other states all reach the same z*.
- **Non-normal**: σ_{max} = 3.2 (median) > 1 > ρ, so perturbations can grow before they decay.

Chasing a moving attractor

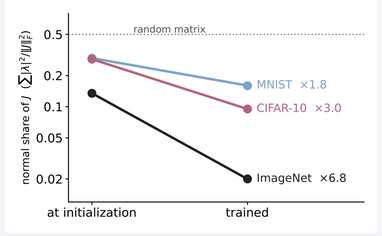


- z* (star) moves along a curve (red); the state (dot) is **5.9** from the previous fixed point, **41.4** from the current.
- The lag is structural: one step shrinks the distance by 0.11 (ρ ≈ 0.14).

Amplification is learned

	ρ	σ_{max}/ρ	feed-fwd
untrained	0.05	7	0.93
random core	0.24	7	0.93
trained	**0.15**	**23**	**0.99**

***random core**: trained image encoder, recurrence at initialization.*

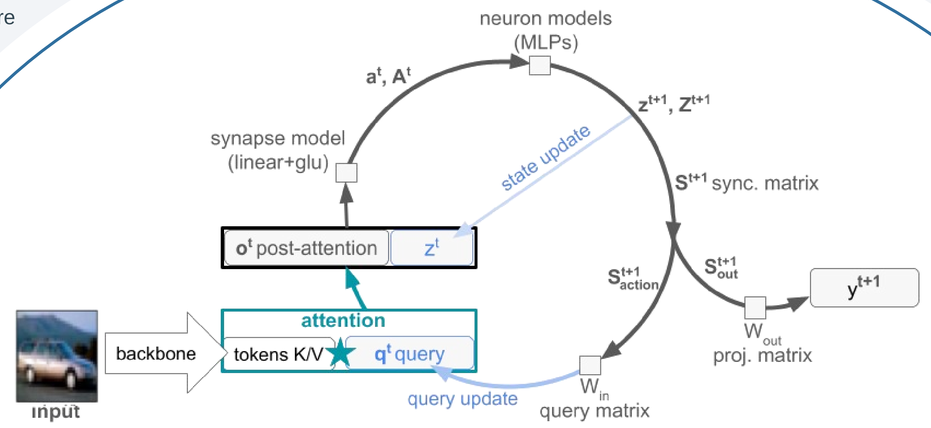


Every CTM is attracting and largely feed-forward at initialization. Training sharpens this into **balanced amplification** (Murphy & Miller, 2009), most on ImageNet, the only model that amplifies transiently (σ_{max} > 1).

The CTM's thought is a curve of attractors that every image follows, one tick behind.

Freeze the memory within a tick: one tick becomes an ordinary map $z^{t+1} = f(z^t; A^t)$ whose fixed point z* we can find, linearize and track.

One tick of the CTM



The image is read **once**. Then the state loops: attend tokens with a query from memory, pass through synapse and neuron models, update the state, accumulate co-activations into a **synchronization memory**, and read the class from it.

model	acc. ↑	ρ(J) med	σ_{max} med / max	attn. spread	frozen q	shuffled q
MNIST (D = 128)	0.95	0.10	0.49 / 0.77	0.97	≈0%	≈0%
CIFAR (ResNet-9)	0.84	0.068	0.36 / 3.06	0.90	+0.3%	+0.2%
ImageNet (D = 4096)	**0.72**	**0.15**	**3.2 / 74**	**0.86**	**−34%**	**−7%**

Attraction (ρ < 1) is universal; amplification (σ_{max} > 1) and query dependence emerge only on ImageNet.
Attention spread: entropy of head-averaged attention over patches, normalized (1 = uniform)
Frozen q: accuracy change with the query *frozen* at its tick-1 value
Shuffled q: query *shuffled* from another image.

Takeaways

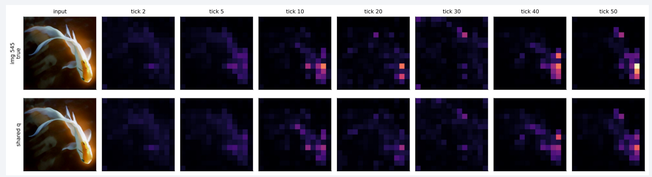
1. Thought is a curve of moving attractors, followed one tick behind.
2. Attention shifts the shared curve; it does not drive it.
3. Training builds the amplification, not the attraction.

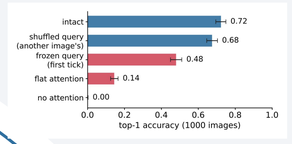
***Open question:** what shapes the curve and its wave?*

References. Darlow, Regan, Risi, Seely & Jones (2025) Continuous Thought Machines · Murphy & Miller (2009) Balanced amplification · Sussillo & Barak (2013) Opening the black box · Trefethen & Embree (2005) Spectra and Pseudospectra.

arXiv:2610.19220

Attention follows a shared schedule

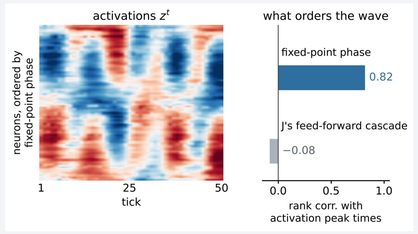




Swapping queries between images costs only **7%**; freezing them, 34%.

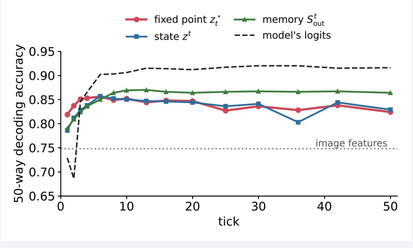
An image-independent increment predicts the next query (**R^2 = 0.98**).

Image-induced traveling wave



- Sorted by z* phase, neurons show a traveling pattern resembling the CTM authors' wave; this order matches activation phase (**0.82**), J's cascade does not.
- Peaks are staggered across the whole thought (s.d. 15.0 of 50 ticks).

Class rides the curve



- From tick 1 the fixed point carries class (**0.82**) above the image features (0.75); the state follows a tick later.
- Shared curve: **66%** of variance. Class: a small sideways shift (**10%**; 0.7% shuffled).